SULLIVAN & McLAUGHLIN
ELECTRICAL CONTRACTORS, INC.
74 Lawley Street
Boston, Ma 02122-3608

PHONE (617) 474-0500
FAX (617) 474-0555 or (617) 268-8385

REQUEST FOR INFORMATION

SULLYMAC RFI # SMC-065 **DATE** 04/01/24

PROJECT NAME: MGH-4B CSP **JOB #** 120048TUR

TO: TURNER & WALSH CONSRUCTION

L5 BALCONY RAILING LIGHTNING PROTECTION

RESPONSE REQUIRED

REQUEST

The L5 balcony requires lightning protection. Attached is a review of submittal 057300-002-1 LIVERS GLASS RAIL SHOP DRAWINGS and a letter from a lightning protection vendor. Please advise if the L5 railing can be modified as suggested in the attached documents.

SIGNED

ANDREW RYAN

BOSTON LIGHTNING ROD COMPANY, INC.
ESTABLISHED 1873 - INCORPORATED1911
MANUFACTURERS-ENGINEERS-INSTALLERS
Listed Conductors & Surge Protectors

April 1, 2024

Sullivan & McLaughlin Co.
74 Lawley Street
Boston, Ma. 02122
Attn: Andrew Ryan

Re: Massachusetts General Hospital
Cambridge Street Project
Boston, Ma.

Subject: Request for Information/Clarification

Mr. Ryan, Please submit the following item for clarification as discussed.

The project has a glass railing system proposed for the 5th floor exterior balcony.

Level 5 is not located in a zone of protection and will require lightning protection to be installed on the railing system for system compliance per UL 96A code requirements.

A review of the proposed railing system per the Livers Bronze shop drawings, shows a glass panel system with no vertical metal support posts. It has a proposed metal cap which is approx.1/8 in metal thickness.

The railing system will need to be redesigned so as not to have air terminals installed on the cap system with interconnecting conductors.

The redesign of the railing system must include a metal cap system with a metal thickness not less than 3/16. The metal cap joints will need to be welded together to make it electrically continuous. The cap system shall be connected to vertical support posts with a metal thickness of not less than 3/16. The vertical posts can then provided with a UL listed bond connection at the base of the posts and terminate onto the roof top lightning protection system installed below the roof top paver system.

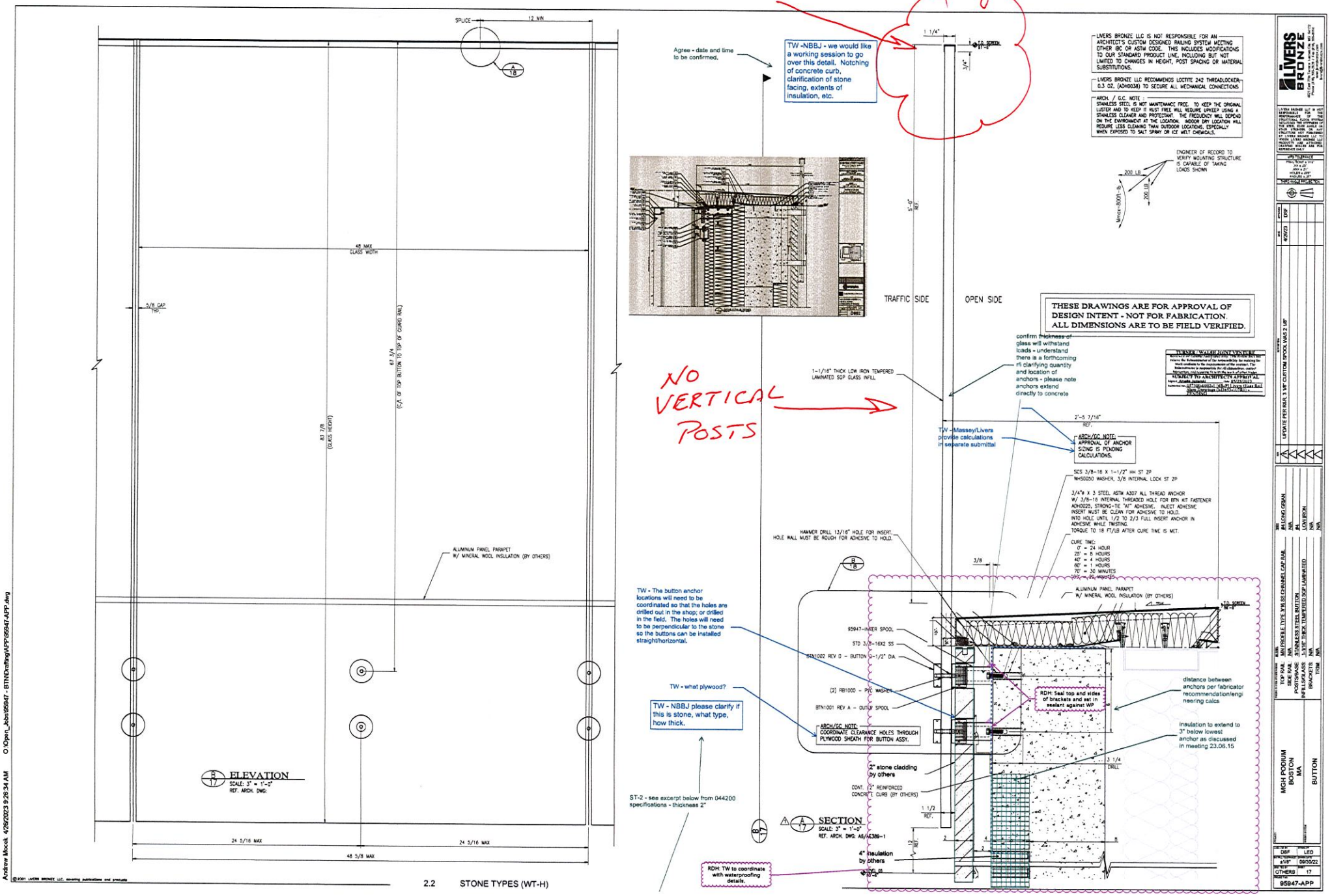
If you have any question's please feel free to contact me.

Sincerely
Boston Lightning Rod Co. Inc.

Andrew Mulholland

WAREHOUSE AND OFFICE: 1201 EAST STREET• DEDHAM, MA• TEL: 781-326-2807• 800-992-3466 FAX: 781-329-3853
BRANCH OFFICE: DENNIS, MA• TEL: 508-398-0222• WWW.BLRCO.NET

RAILING CAP


SPLICE
12 MIN
48 MAX GLASS WIDTH
5/8 GAP TYP.
83 7/8 (GLASS HEIGHT)
87 3/4 (E/L OF TOP BUTTON TO TOP OF GUARD RAIL)
ALUMINUM PANEL PARAPET W/ MINERAL WOOL INSULATION (BY OTHERS)
24 5/16 MAX
24 5/16 MAX
48 5/8 MAX
ELEVATION
SCALE: 3" = 1'-0"
REF. ARCH. DWG:
Agree - date and time to be confirmed.
TW -NBBJ - we would like a working session to go over this detail. Notching of concrete curb, clarification of stone facing, extents of insulation, etc.
1 1/4"
3/4"
5'-0" REF.
LIVERS BRONZE LLC IS NOT RESPONSIBLE FOR AN ARCHITECT'S CUSTOM DESIGNED RAILING SYSTEM MEETING EITHER IBC OR ASTM CODE. THIS INCLUDES MODIFICATIONS TO OUR STANDARD PRODUCT LINE, INCLUDING BUT NOT LIMITED TO CHANGES IN HEIGHT, POST SPACING OR MATERIAL SUBSTITUTIONS.
LIVERS BRONZE LLC RECOMMENDS LOCTITE 242 THREADLOCKER, 0.3 OZ, (ADH0038) TO SECURE ALL MECHANICAL CONNECTIONS
ARCH. / G.C. NOTE :
STAINLESS STEEL IS NOT MAINTENANCE FREE. TO KEEP THE ORIGINAL LUSTER AND TO KEEP IT RUST FREE WILL REQUIRE UPKEEP USING A STAINLESS CLEANER AND PROTECTANT. THE FREQUENCY WILL DEPEND ON THE ENVIRONMENT AT THE LOCATION. INDOOR DRY LOCATION WILL REQUIRE LESS CLEANING THAN OUTDOOR LOCATIONS, ESPECIALLY WHEN EXPOSED TO SALT SPRAY OR ICE MELT CHEMICALS.
ENGINEER OF RECORD TO VERIFY MOUNTING STRUCTURE IS CAPABLE OF TAKING LOADS SHOWN
200 LB
200 LB
Minor-Roof: lb
TRAFFIC SIDE
OPEN SIDE
THESE DRAWINGS ARE FOR APPROVAL OF DESIGN INTENT - NOT FOR FABRICATION. ALL DIMENSIONS ARE TO BE FIELD VERIFIED.
confirm thickness of glass will withstand loads - understand there is a forthcoming rfi clarifying quantity and location of anchors - please note anchors extend directly to concrete
TURNER - WALSH JOINT VENTURE
SUBJECT TO ARCHITECT'S APPROVAL
1-1/16" THICK LOW IRON TEMPERED LAMINATED SGP GLASS INFILL
NO VERTICAL POSTS
2'-5 7/16" REF.
TW - Massey/Livers provide calculations in separate submittal
ARCH/GC NOTE: APPROVAL OF ANCHOR SIZING IS PENDING CALCULATIONS.
SCS 3/8-16 X 1-1/2" HH ST ZP
WH50050 WASHER, 3/8 INTERNAL LOCK ST ZP
3/4"ø X 3 STEEL ASTM A307 ALL THREAD ANCHOR W/ 3/8-16 INTERNAL THREADED HOLE FOR BTN KIT FASTENER ADH0025, STRONG-TIE "A1" ADHESIVE. INJECT ADHESIVE INSERT MUST BE CLEAN FOR ADHESIVE TO HOLD. INTO HOLE UNTIL 1/2 TO 2/3 FULL INSERT ANCHOR IN ADHESIVE WHILE TWISTING. TORQUE TO 18 FT/LB AFTER CURE TIME IS MET.
CURE TIME:
0° = 24 HOUR
23° = 8 HOURS
40° = 4 HOURS
60° = 1 HOURS
70° = 30 MINUTES
95° = 20 MINUTES
HAMMER DRILL 13/16" HOLE FOR INSERT. HOLE WALL MUST BE ROUGH FOR ADHESIVE TO HOLD.
3/8
TW - The button anchor locations will need to be coordinated so that the holes are drilled out in the shop; or drilled in the field. The holes will need to be perpendicular to the stone so the buttons can be installed straight/horizontal.
ALUMINUM PANEL PARAPET W/ MINERAL WOOL INSULATION (BY OTHERS)
95947-INNER SPOOL
STD 3/8-16X2 SS
BTN1002 REV D - BUTTON 3-1/2" DIA.
TW - what plywood?
(2) RB1000 - PVC WASHER
BTN1001 REV A - OUTER SPOOL
TW - NBBJ please clarify if this is stone, what type, how thick.
ARCH/GC NOTE: COORDINATE CLEARANCE HOLES THROUGH PLYWOOD SHEATH FOR BUTTON ASSY.
RDH: Seal top and sides of brackets and set in sealant against WP
distance between anchors per fabricator recommendation/engineering calcs
insulation to extend to 3" below lowest anchor as discussed in meeting 23.06.15
3 1/4 DRILL
2" stone cladding by others
CONT. 2" REINFORCED CONCRETE CURB (BY OTHERS)
1 1/2 REF.
12 REF.
4" insulation by others
ST-2 - see excerpt below from 044200 specifications - thickness 2"
RDH: TW to coordinate with waterproofing details.
SECTION
SCALE: 3" = 1'-0"
REF. ARCH. DWG: A5/A4309-1
LIVERS BRONZE
UPDATE PER RAIL 3 1/8" CENTER SPOOL WAS 2 1/8"
TOP RAIL: 95947-PROFILE TYPE 3 V16 SS CHANNEL CAP RAIL
SIDE RAIL: N/A
POSTS/BASE: STAINLESS STEEL BUTTON
INFILL/GLASS: 1-1/16" THICK TEMPERED SGP LAMINATED
BRACKETS: N/A
TOTE: N/A
MGH PODIUM
BOSTON
MA
BUTTON
DBF
LEO
1/8"
09/00/22
OTHERS
17
95947-APP
Andrew Mocek 4/26/2023 9:28:34 AM O:\Open_Jobs\95947 - BTN\Drafting\APP\95947-APP.dwg
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2.2 STONE TYPES (WT-H)

EXAMPLE OF VERTICAL POST AND RAILING CAP.